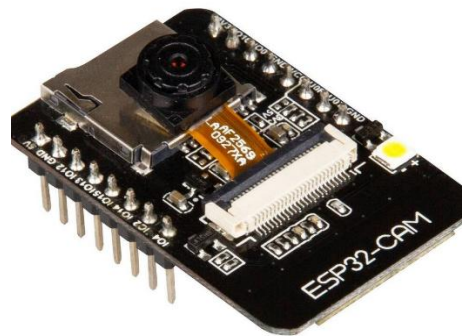


AMERICAN SIGN LANGUAGE DETECTION USING ESP32-CAM AND TINYML

Real-Time Embedded Gesture Recognition using TensorFlow Lite
for Microcontrollers

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ABSTRACT

This project presents a real-time American Sign Language (ASL) recognition system implemented on the ESP32-CAM platform using TinyML techniques. The objective of the system is to recognize static ASL hand gestures directly on-device without relying on cloud computation or external processing hardware. The solution demonstrates the feasibility of deploying lightweight deep-learning models on resource-constrained embedded systems.

The ESP32-CAM captures live image frames using the OV2640 camera module. The captured frames are pre-processed through grayscale conversion, resizing, and down sampling to generate a compact 28×28 input suitable for inference. A quantized TensorFlow Lite Micro Convolutional Neural Network (CNN) model is used for gesture classification. The model is optimized for low memory consumption and real-time performance on the ESP32 microcontroller.

The complete inference pipeline, including preprocessing, model execution, and output generation, runs entirely on-device. The predicted ASL gesture is displayed through Serial output and a lightweight HTTP web interface hosted by the ESP32-CAM. The system achieves real-time performance while maintaining efficient RAM and Flash memory utilization.

This project highlights the potential of TinyML in enabling low-cost, portable, and energy-efficient edge-AI applications for assistive communication systems and embedded intelligence.

INTRODUCTION

American Sign Language (ASL) is one of the most widely used sign languages for communication among hearing-impaired individuals. However, communication barriers still exist between sign-language users and people unfamiliar with ASL. Recent advances in Embedded AI and TinyML provide opportunities to develop compact and affordable systems capable of performing intelligent gesture recognition directly on low-power hardware.

This project focuses on developing a real-time ASL gesture recognition system using the ESP32-CAM microcontroller and TensorFlow Lite for Microcontrollers (TFLM). The system is capable of recognizing 24 static ASL alphabet gestures from A to Y, excluding J and Z due to their dynamic motion-based nature.

The primary objective of the project is to demonstrate that deep-learning inference can be successfully executed on resource-constrained embedded hardware while maintaining acceptable real-time performance. By performing all computation locally on the ESP32-CAM, the system eliminates the need for cloud connectivity, making it portable, low-latency, and suitable for edge-AI applications.

The project combines concepts from Embedded Systems, Computer Vision, Machine Learning, and TinyML to create a practical and efficient assistive technology solution.

PROBLEM STATEMENT

Communication between hearing-impaired individuals and people unfamiliar with sign language remains a major challenge in daily life. Most existing gesture-recognition systems depend on high-performance computing devices or cloud-based processing, which increases cost, power consumption, and latency.

The objective of this project is to develop a compact, low-cost, and portable ASL recognition system capable of performing real-time gesture classification entirely on embedded hardware. The proposed system aims to provide an efficient TinyML-based solution using ESP32-CAM while maintaining acceptable accuracy and real-time performance.

OBJECTIVES

- To develop a real-time ASL gesture recognition system using ESP32-CAM.
 - To implement TinyML-based inference on resource-constrained hardware.
 - To preprocess camera frames into lightweight grayscale input suitable for embedded inference.
 - To deploy a quantized TensorFlow Lite Micro CNN model on ESP32.
 - To achieve efficient memory utilization and low inference latency.
 - To provide gesture predictions through Serial Monitor and HTTP web interface.
 - To demonstrate the feasibility of Edge AI for assistive communication systems.
-

APPLICATIONS

- Assistive communication systems
 - Smart human-computer interaction
 - Embedded AI systems
 - Edge-based gesture recognition
 - IoT-based intelligent devices
 - Educational TinyML demonstrations
 - Real-time embedded vision systems
-

SYSTEM OVERVIEW

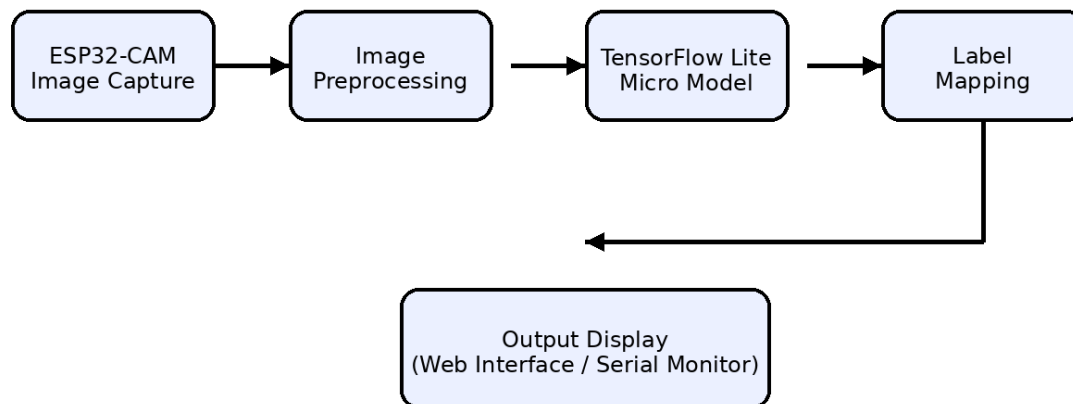
The system captures image frames using the OV2640 camera module integrated with the ESP32-CAM. The captured frames are resized and converted into grayscale images before being passed to the TensorFlow Lite Micro inference engine.

The complete processing pipeline includes:

1. Image Capture using ESP32-CAM
2. Image Preprocessing and Downsampling
3. TensorFlow Lite Micro Inference
4. Probability Calculation and Label Mapping
5. Output Display through Serial Monitor/Web Interface

The entire pipeline executes directly on the ESP32-CAM without requiring external processing hardware or cloud computation.

System Architecture Diagram

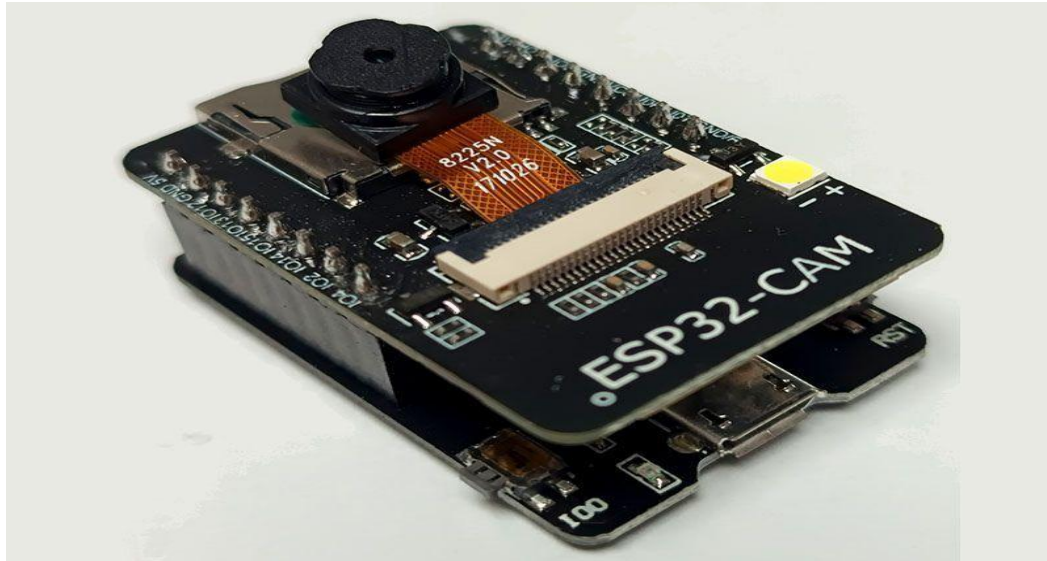


HARDWARE COMPONENTS

ESP32-CAM

The ESP32-CAM is a low-cost embedded development board featuring:

- ESP32 dual-core microcontroller
- OV2640 camera module
- WiFi capability



- PSRAM support for image processing tasks

The board serves as both the image acquisition device and the inference platform.

OV2640 Camera Module

The OV2640 camera captures live image frames that are processed for ASL gesture recognition.



USB-to-Serial Programmer

A USB-to-Serial module is used to upload firmware to the ESP32-CAM board during development.



Power Supply

The system operates using a regulated 5V power source through USB.

SOFTWARE TOOLS USED

Tool	Purpose
Arduino IDE	ESP32 firmware development
TensorFlow Lite Micro	TinyML inference engine
Python	Model training and preprocessing
OpenCV	Image preprocessing
ESP32 Board Package	ESP32 compilation support
GitHub	Source code management

DATASET & PREPROCESSING

The model is trained on a 24-class ASL alphabet dataset containing static hand gestures corresponding to the letters A–Y, excluding J and Z. The dataset contains multiple gesture samples captured under different lighting conditions and hand orientations.

To improve model robustness, preprocessing and augmentation techniques are applied:

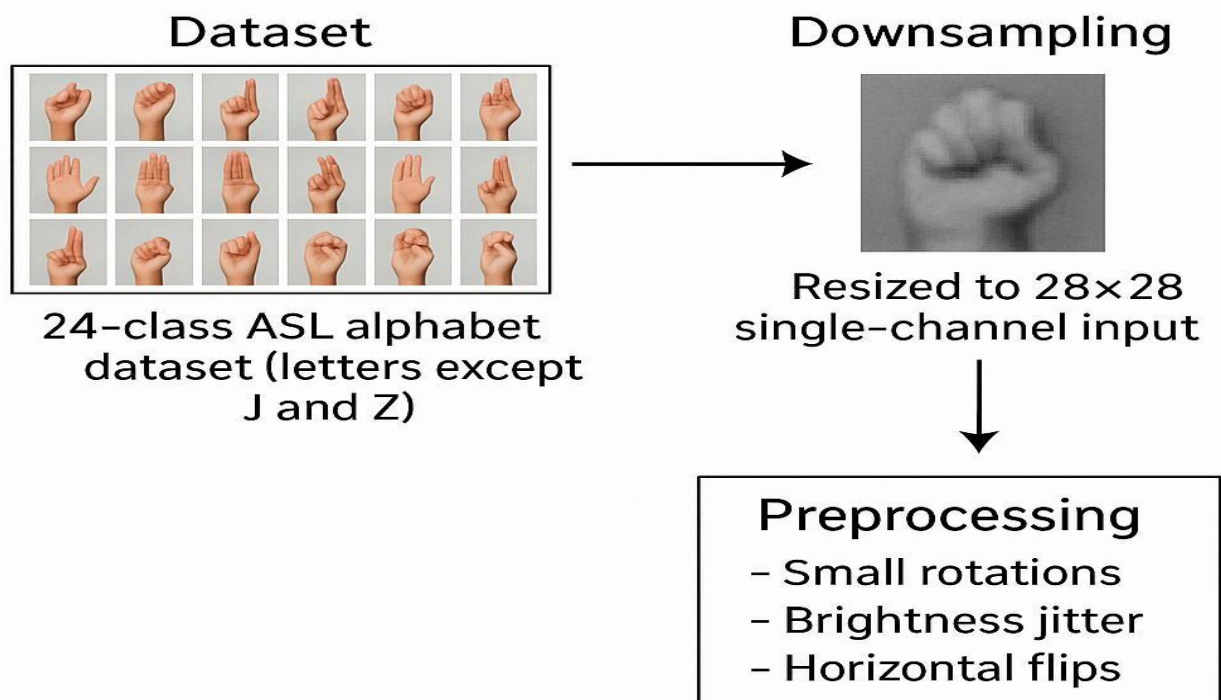
- Grayscale conversion

- Image resizing to 28×28
- Brightness adjustment
- Rotation augmentation
- Horizontal flipping

The ESP32-CAM captures RGB565 frames which are downsampled and converted into grayscale format before inference.



Dataset & Preprocessing



MODEL ARCHITECTURE

The project uses a lightweight Convolutional Neural Network (CNN) optimized for TinyML deployment. The model is designed specifically for low-memory embedded hardware.

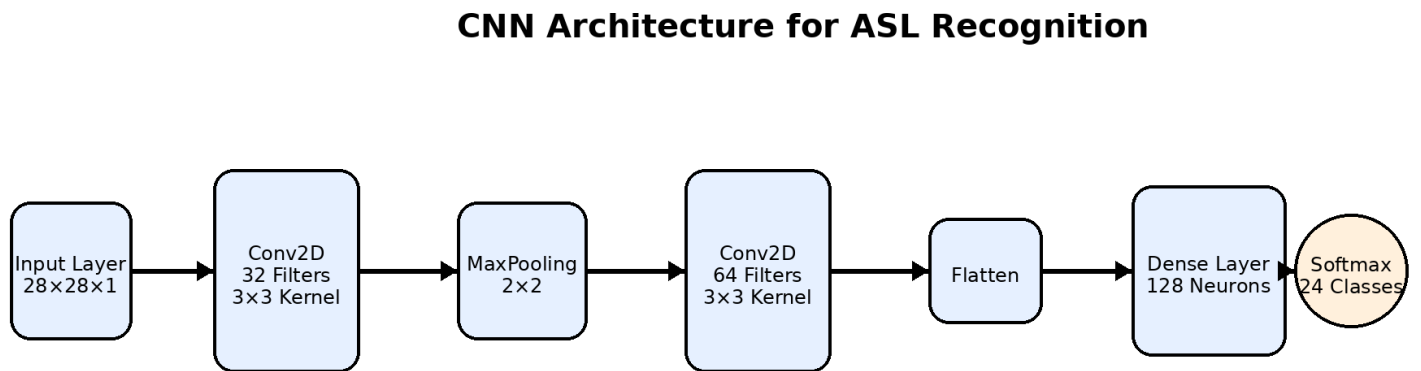
The CNN architecture includes:

- Convolutional layers

- Pooling layers
- Activation functions
- Fully connected output layer
- Softmax classification

The model accepts 28×28 grayscale input images and outputs probabilities for 24 ASL gesture classes.

Figure 1: System Architecture Diagram



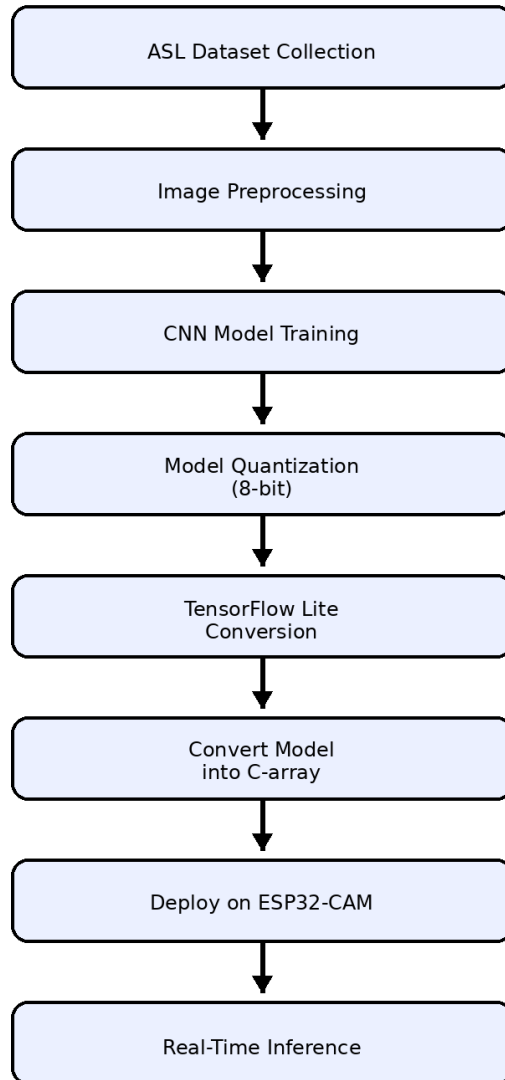
TINYML OPTIMIZATIONS

Several optimizations were implemented to ensure efficient execution on the ESP32-CAM:

- 8-bit quantized TensorFlow Lite Micro model
- Static tensor arena allocation
- Reduced image resolution
- Lightweight preprocessing pipeline
- Operator pruning for required TensorFlow operations only
- Efficient memory management

These optimizations significantly reduce Flash and RAM usage while improving inference speed.

TinyML Deployment Pipeline



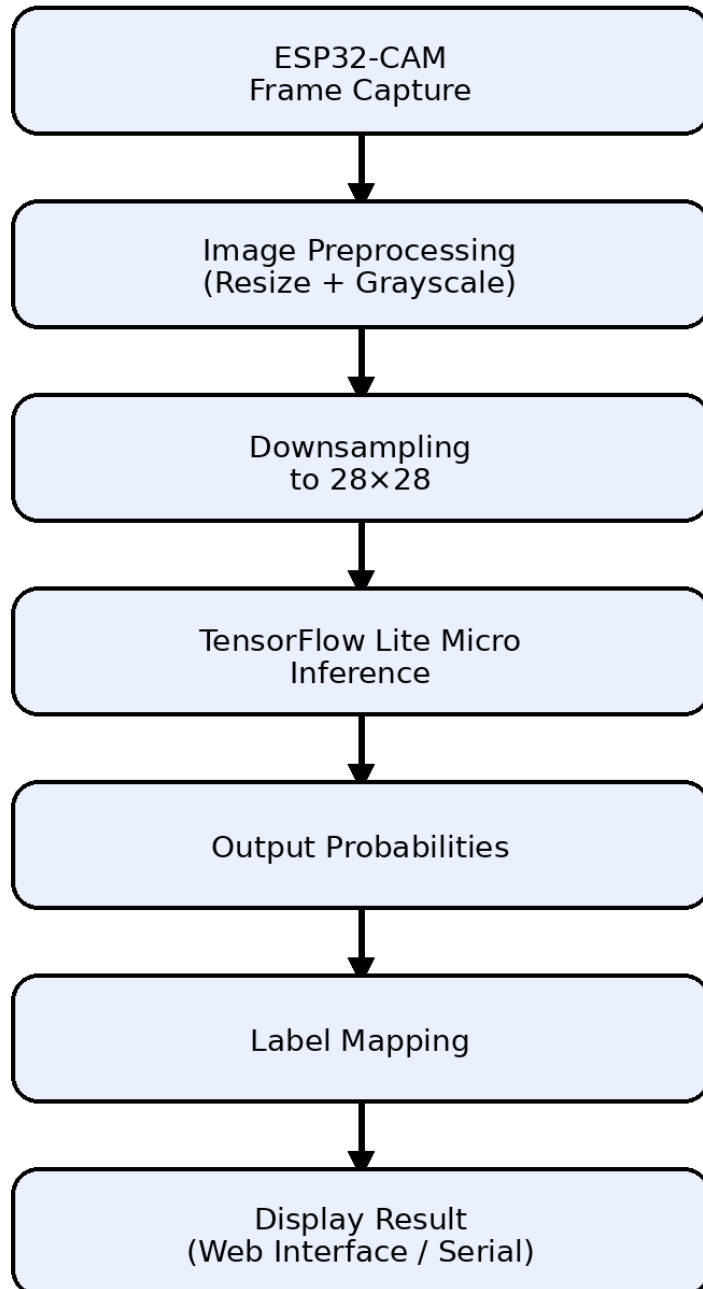
SYSTEM WORKFLOW

The complete workflow of the proposed system is:

1. ESP32-CAM captures image frame
2. Frame preprocessing and resizing
3. Conversion to grayscale image
4. TensorFlow Lite Micro inference

5. Output probability generation
6. Label mapping
7. Display recognized ASL gesture

ASL Recognition System Workflow



DEPLOYMENT ON ESP32-CAM

The trained TensorFlow Lite model is converted into a C-array and embedded directly into the ESP32 firmware. During runtime, the ESP32-CAM captures image frames, preprocesses them, and performs inference using TensorFlow Lite for Microcontrollers.

The output probabilities are mapped to corresponding ASL labels, and the final prediction is displayed through:

- Serial Monitor
- HTTP Web Interface

The system operates completely offline and performs all computation on-device.

HARDWARE SETUP & DEPLOYMENT

The hardware setup includes the ESP32-CAM module connected with a USB-to-Serial programmer for firmware uploading and debugging. The deployed TinyML model runs directly on the ESP32 microcontroller.

EXPERIMENTAL RESULTS

The implemented system successfully recognizes static ASL gestures in real time using the ESP32-CAM.

The prediction output is displayed through the hosted web interface and Serial Monitor. Multiple gesture classes were tested under different lighting conditions and hand orientations.

ESP32 ASL Sign Detection

Not secure 10.219.140.183

Summarize

Sign in

Chat

ASL Detection (ESP32-CAM)



Detect Sign

Done in 3862 ms

Top1: Y, Top2: A

17°C Clear

Search

ENG IN

23:22 21-11-2025

ESP32 ASL Sign Detection

Not secure 10.219.140.183

Summarize

Sign in

Chat

ASL Detection (ESP32-CAM)



Detect Sign

Done in 3863 ms

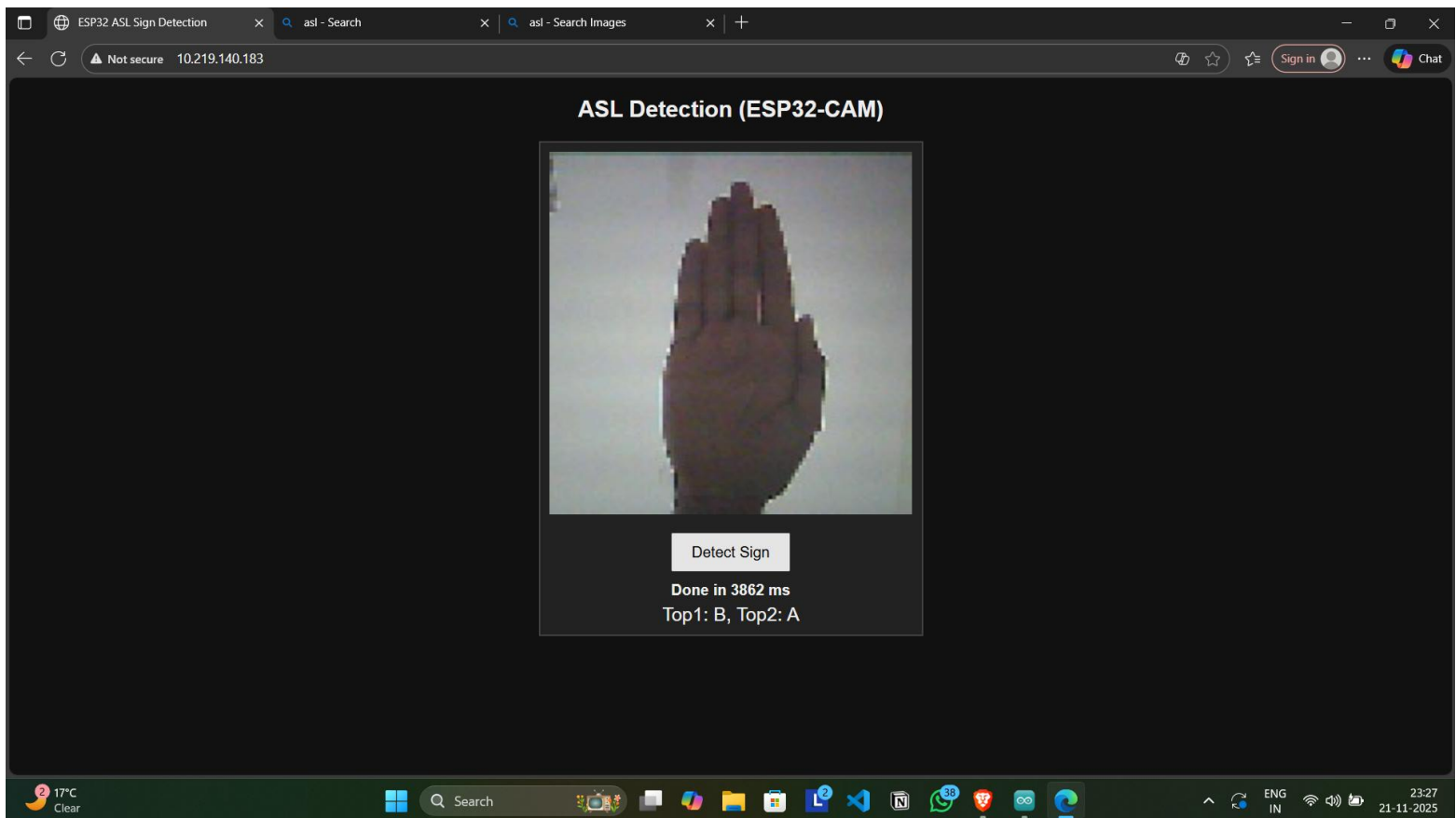
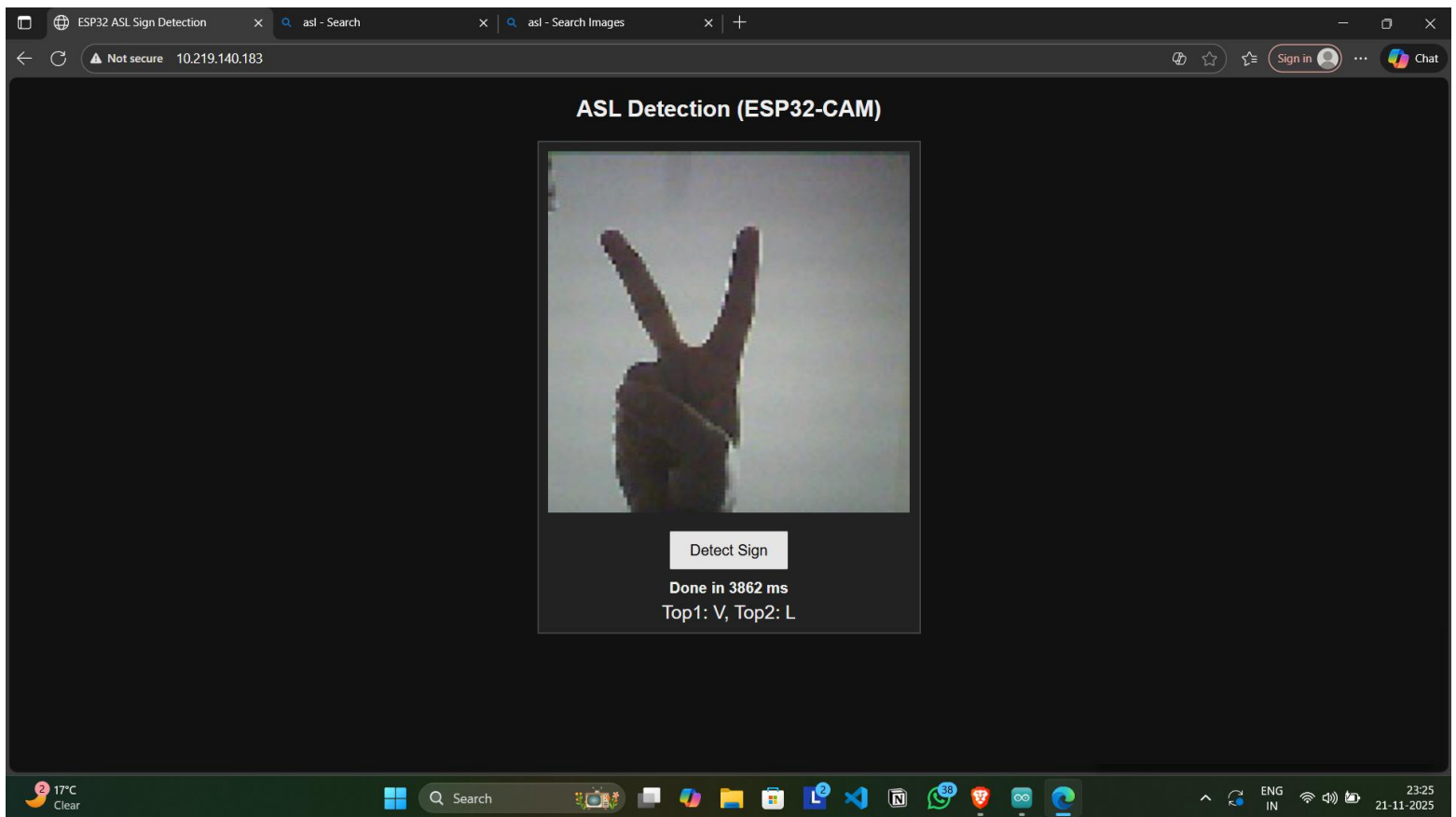
Top1: L, Top2: A

Trending videos The Day My Do...

Search

ENG IN

23:21 21-11-2025

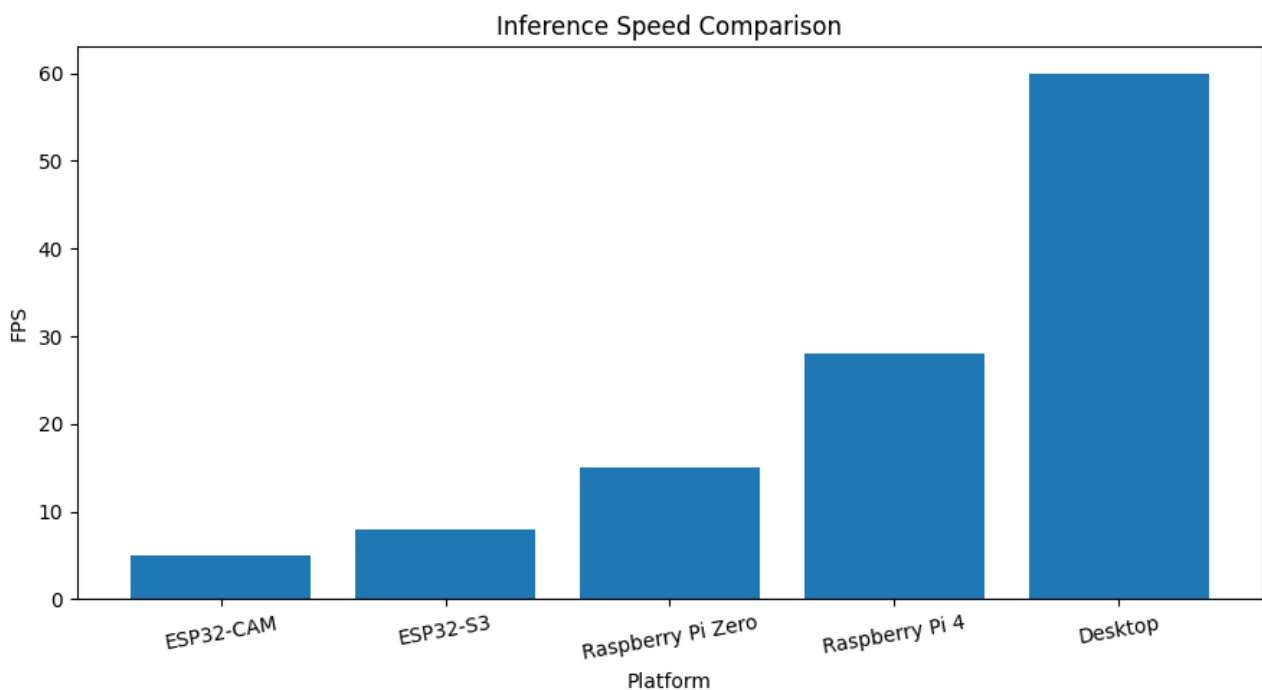


PERFORMANCE ANALYSIS

Observed Performance

Parameter	Value
Test Accuracy	~87–90%
Inference Time	~45–60 ms
Real-time FPS	~5 FPS
Model Flash Size	~520 KB
Tensor Arena Size	~12–16 KB
Working RAM Usage	~180–220 KB

The results demonstrate that TinyML models can achieve reliable real-time inference on low-cost embedded hardware.



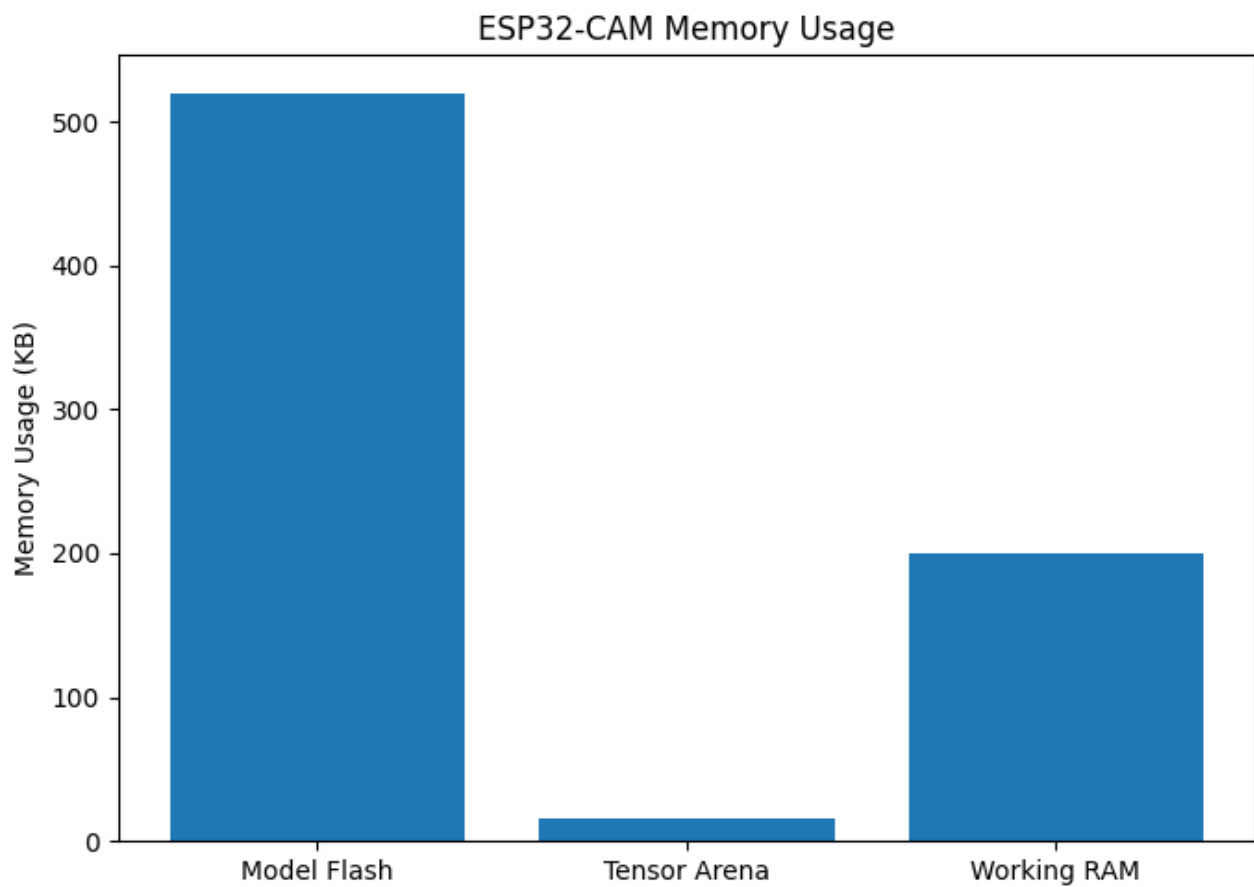
MEMORY USAGE & OPTIMIZATION

The ESP32-CAM has limited memory resources; therefore, efficient memory optimization is essential for successful TinyML deployment.

The optimized model utilizes:

- Flash memory for storing the quantized TensorFlow Lite model
- Tensor Arena for intermediate ML computations
- RAM for camera buffers and preprocessing operations

Static memory allocation and quantization significantly reduce memory consumption while maintaining stable inference performance.

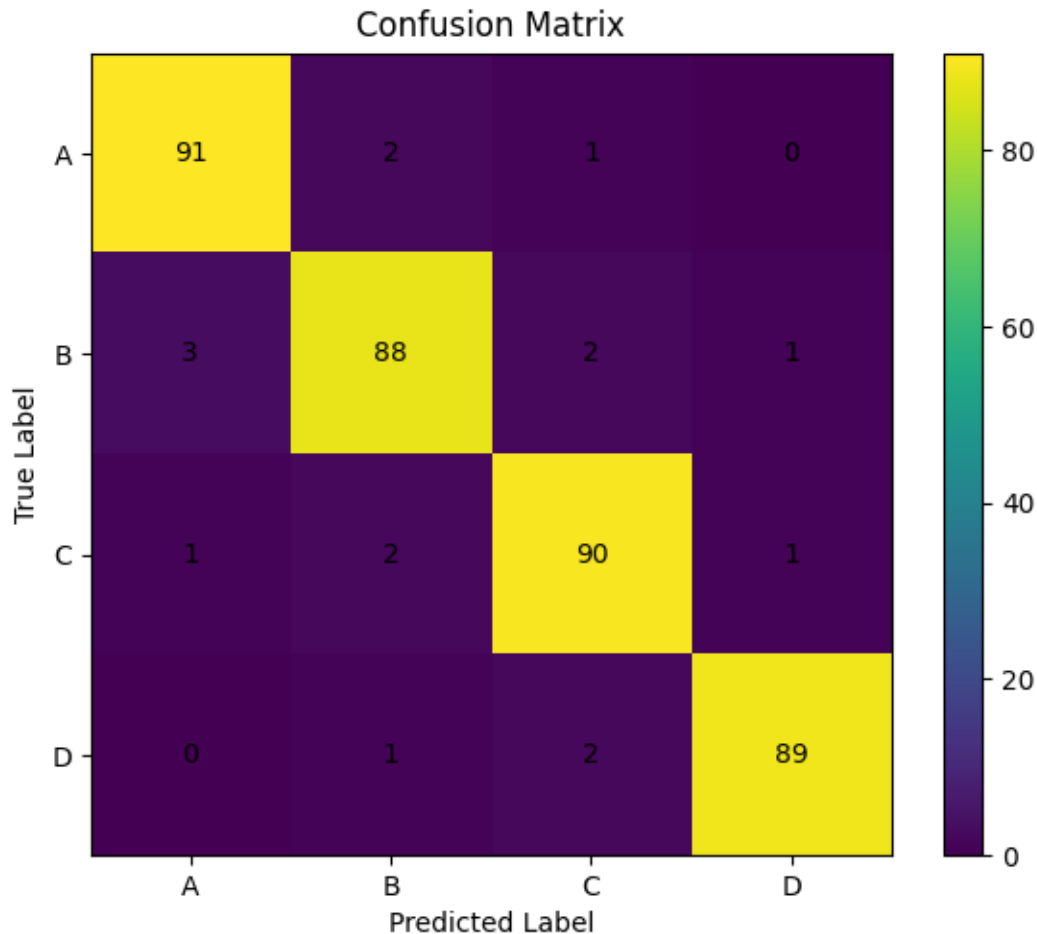


CONFUSION MATRIX ANALYSIS

The confusion matrix illustrates the classification performance of the TinyML model across multiple ASL gesture classes.

Higher values along the diagonal indicate correct predictions, while off-diagonal values represent misclassifications. The results demonstrate strong class separation and reliable gesture recognition performance.

The overall classification accuracy achieved by the model is approximately 87–90%.



ADVANTAGES OF THE SYSTEM

- Low-cost implementation
- Real-time on-device inference
- No cloud dependency
- Portable and energy efficient
- Fast response time
- TinyML deployment on constrained hardware
- Easy integration with IoT systems

CHALLENGES FACED

- Limited RAM availability on ESP32-CAM
 - Optimizing inference speed for real-time execution
 - Memory management during image preprocessing
 - Model size reduction for embedded deployment
 - Maintaining accuracy after quantization
 - Lighting variation during gesture detection
-

LIMITATIONS

- Supports only static ASL gestures
 - Lower FPS compared to high-end processors
 - Accuracy affected by poor lighting conditions
 - Limited computational resources of ESP32-CAM
 - Small input image resolution
-

FUTURE SCOPE

- Support for dynamic gestures such as J and Z
 - Integration of audio/text output
 - Improved CNN architectures for higher accuracy
 - Mobile application integration
 - Deployment on advanced TinyML hardware such as ESP32-S3
 - Multi-hand gesture recognition
 - Edge-AI optimization using hardware accelerators
-

CONCLUSION

This project demonstrates the successful implementation of a TinyML-based American Sign Language recognition system on the ESP32-CAM platform. The system achieves real-time gesture classification while operating entirely on-device using a quantized TensorFlow Lite Micro model.

The project highlights the capabilities of Embedded AI and TinyML for developing low-cost, portable, and efficient assistive technologies. Despite the limited computational resources of the

ESP32-CAM, the system delivers reliable inference performance through optimized preprocessing, quantization, and memory management techniques.

The proposed solution demonstrates that powerful AI applications can be deployed successfully on low-power embedded systems, opening new opportunities for edge computing and smart assistive technologies.

REFERENCES

1. TensorFlow Lite for Microcontrollers Documentation
2. ESP32-CAM Technical Reference Manual
3. Arduino ESP32 Development Framework
4. TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers
5. OpenCV Documentation
6. American Sign Language Dataset Sources
7. TensorFlow Official Documentation

APPENDIX (SOURCE CODE)

The complete source code implementation, including:

- Main ESP32 firmware
- TensorFlow Lite model integration
- Camera interface
- Downsampling functions
- Web interface implementation
- Model configuration files

is attached separately and available through the project GitHub repository.

GitHub Repository Link: <https://github.com/0xKhushwant/tinyml-asl-recognition-esp32cam>